4 Functions: representations and reference functions

Sheet 19 Thu 3 Dec. 2026 *Real-valued function; domain of definition*

1. Warm-up · Quick Solve $x - 4 = 0$ then $2x + 6 = 0$.

2. Core · Method Domain of definition of $f(x) = \dfrac{1}{x-4}$?

3. Core + · MCQ The domain of definition of $h(x) = \dfrac{1}{x^2+1}$ is:

A. $\mathbb{R} \setminus \{1\}$ **B.** $\mathbb{R} \setminus \{-1, 1\}$ **C.** $\mathbb{R}$ **D.** $\mathbb{R}^*$

4. Review · Table Is this table a proportionality table?

x	2	5	8
y	6	15	24

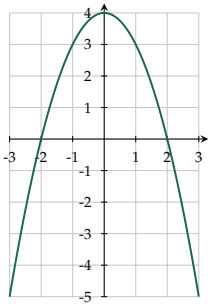
5. Thread · Code reading What does $f(4)$ return?

```
def f(x):
    return 3*x - 2
```

Sheet 20 Mon 7 Dec. 2026 *Curve with equation $y = f(x)$; images and preimages*

1. Warm-up · Quick $f(x) = -x^2 + 4$. Calculate $f(0)$, $f(-2)$ and $f(3)$.

2. Core · Graph Curve of f:



Read off $f(1)$ and the preimages of 0.

3. Core + · Thinking For this function, does the equation $f(x) = 5$ have any solutions?

4. Review · Quick Urn: 5 red balls, 3 blue, 2 green. Probability of drawing a blue one? A non-red one?

5. Thread · Method Solve in $\mathbb{R}$: $4 - 3x = 10$.

Sheet 21 Tue 8 Dec. 2026 *Reference functions: absolute value, square, inverse*

1. Warm-up · Quick Calculate $|-2.5|$, $(-3)^2$ and $\dfrac{1}{-4}$.

2. Core · Table Complete, for $x = -2\,;\,-1\,;\,0\,;\,1\,;\,2$, the values of $x \mapsto |x|$, $x \mapsto x^2$ and $x \mapsto \dfrac{1}{x}$.

3. Core + · MCQ Which reference curve passes through $(-1, 1)$, $(0, 0)$, $(1, 1)$ and $(2, 4)$?

A. the line $y = x$ **B.** the inverse function **C.** the absolute value function **D.** the square function

4. Review · Quick An object is enlarged with scale factor 1.5. What does a length of 8 cm become? By what number is the area multiplied?

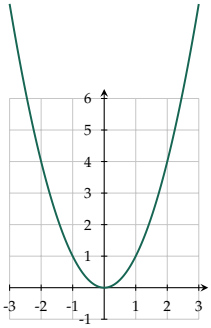
5. Thread · Prove Conjecture: "the sum of two consecutive integers is odd". Prove it.

Sheet 22 Thu 10 Dec. 2026 *Square root and cube; sign of reference functions*

1. Warm-up · Quick Calculate $\sqrt{49}$, $\sqrt{0.25}$ and $(-2)^3$.
2. Core · Method Study the sign of $f(x) = -2x + 6$.
3. Core + · True/false True or false? Justify. For every real number x, $\sqrt{x^2} = x$.
4. Review · Quick How many squares make up a net of a cube? Does a cross (a row of 4 squares + 1 square on each side) work?
5. Thread · Quick How many faces, vertices and edges does a square-based pyramid have?

Sheet 23 Mon 14 Dec. 2026 *Graphical and numerical solving: $f(x) = k$, $f(x) < k$*

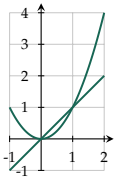
1. Warm-up · Quick Solve $x^2 = 16$ then $x^2 = -4$.
2. Core · Graph Curve of $f(x) = x^2$:



   Solve graphically $f(x) = 4$ then $f(x) < 4$.
3. Core + · Method Solve $|x| < 3$ then $|x| = 5$.
4. Review · Pitfall Calculate $-x^2$ then $(-x)^2$ for $x = -3$.
5. Thread · Method In a triangle ABC, M and N are the midpoints of $[AB]$ and $[AC]$. What is being proved? Which property is cited?

Sheet 24 Tue 15 Dec. 2026 *Solving $f(x) = g(x)$ and $f(x) < g(x)$*

1. Warm-up · Quick Solve $2x - 1 = x + 3$.
2. Core · Graph Curves of $f(x) = x$ and $g(x) = x^2$:



   Solve $f(x) = g(x)$ then $f(x) < g(x)$.
3. Core + · Code reading What does `f(-1) == g(-1)` return? And `f(0) == g(0)`?

```
def f(x):
    return x*x
def g(x):
    return 2*x + 3
```

4. Review · Problem Plan A: 15,000 Ar fixed + 1,000 Ar per session. Plan B: 2,500 Ar per session. From how many sessions is A cheaper?
5. Thread · Quick Calculate the coordinates of the midpoint of $[AB]$ with $A(-2, 5)$ and $B(6, 1)$.